

Use of force-controlled grippers

What you will learn in this chapter

How to use myBlockly to control the force-controlled gripper attached to the myCobot 320 M5Stack robot arm

Some API displays

- **Module 1:** Set Pro Force Control Claw Parameters

Set Pro Force Control Claw Parameters ID 14 | Address 3 | Value 0

- **Applicable scope:** ER myCobot 320 series, ER Mercury series, ER myCobot Pro 600, ERmyCobot Pro 630, other general robots
- **Function:** Set the parameters of the Pro force-controlled gripper. You can set a variety of parameter functions. Please see the following table for details.
- **parameter:**
 - Gripper ID: gripper_id, the default value is 14, and the value range is 1 ~ 254.
 - The command number of the gripper: address.
 - Parameter value: value.

Function	gripper_id	address	value
Set the gripper ID	14	3	1 ~ 254
Set the gripper enable state	14	10	0 or 1, 0 - disable; 1 - enable
Set the clockwise runnable error of the gripper	14	21	0 ~ 16
Set the anti-clockwise runnable error of the gripper	14	23	0 ~ 16
Setting the minimum actuation force of the gripper	14	25	0 ~ 254
IO output settings	14	29	0, 1, 16, 17
Set the IO opening angle	14	30	0 ~ 100
Set IO closing angle	14	31	0 ~ 100
Set the virtual position value of the servo	14	41	0 ~ 100
Setting the holding current	14	43	1 ~ 254

- **Module 2:** Set Pro Force Control Claw Angle

Set Pro Force Control Claw Angle ID 14 | Angle 0

- **Applicable scope:** ER myCobot 320 series, ER Mercury series, ER myCobot Pro 600, ERmyCobot Pro 630, other general robots
- **Function:** Set the angle of the force-controlled gripper.
- **parameter:**
 - Gripper ID: Default is 14, value range is 1 ~ 254.
 - Gripper angle: value range is 0 ~ 100.

- **Module 3:** Set Pro Force Control Claw Torque

Set Pro Force Control Claw Torque ID 14 | Torque value 0

- **Applicable scope:** ER myCobot 320 series, ER Mercury series, ER myCobot Pro 600, ERmyCobot Pro 630, other general robots
- **Function:** Set the torque of the force-controlled gripper.
- **parameter:**
 - Gripper ID: Default is 14, value range is 1 ~ 254.
 - Torque value: range is 1 ~ 100.

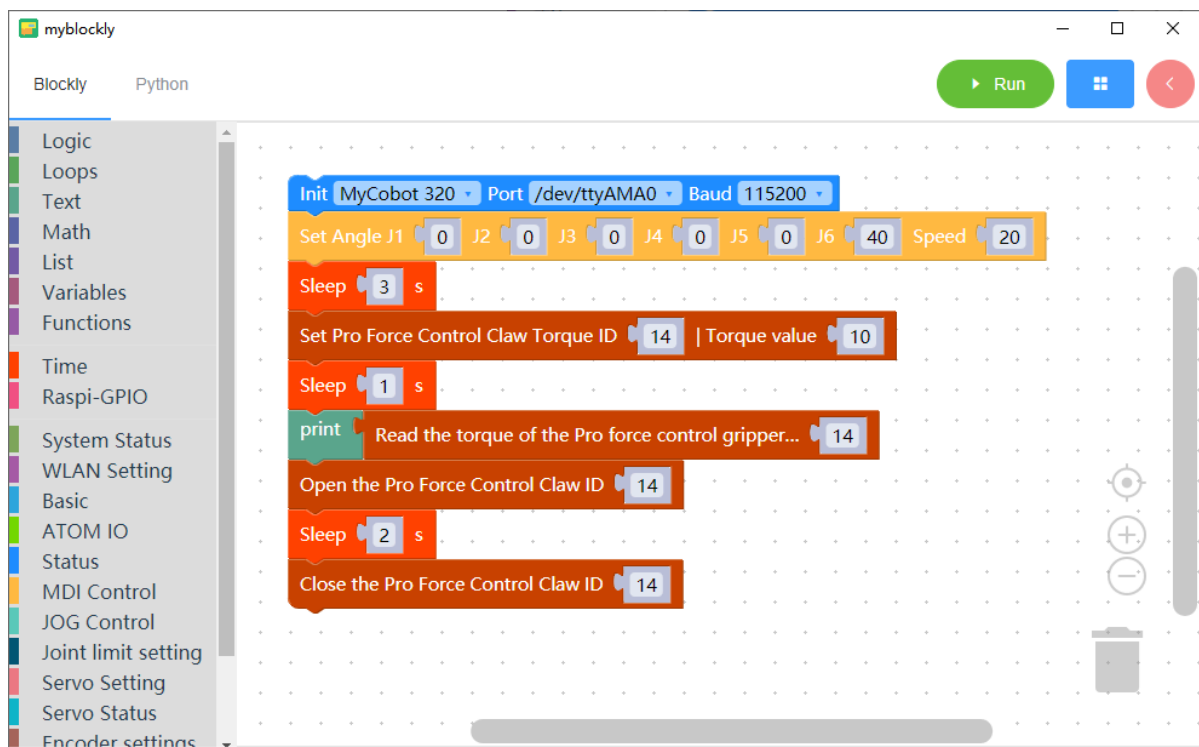
- **Module 4:** Read the torque of the Pro force control gripper

Read the torque of the Pro force control gripper... 14

- **Applicable scope:** ER myCobot 320 series, ER Mercury series, ER myCobot Pro 600, ERmyCobot Pro 630, other general robots
- **Function:** Read the torque of the force-controlled gripper.
- **parameter:**
 - Gripper ID: Default is 14, value range is 1 ~ 254.

Simple Demo

- The graphics code is as follows:



- Implementation content:

The robot arm's joint 6 moves to 40 degrees at a speed of 20, and the rest of the joints move to 0 degrees. After three seconds,

Set the torque of the force-controlled gripper to 10. After one second,

The program running window will print out the current torque value of the force-controlled gripper, then open the gripper. After two seconds,

The gripper closes and the program ends.